Comp Method's Hw # 5

1.)

$$A_1 = \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$$

Find L first:

$$\begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix} \xrightarrow{R2 - 2R1 \to R1}$$

$$\begin{pmatrix} 2(-2) & (1)(-2) \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} -5 & 0 \\ -1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} -5 & 0 \\ -1 & 2 \end{pmatrix} \xrightarrow{\frac{1}{2}R_2 \to R_1} \begin{pmatrix} -5 & 0 \\ -\frac{1}{2} & 1 \end{pmatrix} \xrightarrow{-\frac{1}{5}R_1 \to R_1}$$

$$= \begin{pmatrix} 1 & 0 \\ -\frac{1}{2} & 1 \end{pmatrix} = L$$

U - matrix :

$$A_1 = \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix} \xrightarrow{R1 + 2R2 \to R2}$$

$$= \begin{pmatrix} 2 & 1 \\ -1(2) & 2(2) \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 1 \\ 0 & 5 \end{pmatrix} \xrightarrow{\frac{1}{2}R1 \to R1,\ \frac{1}{5}R2 \to R2}$$

$$=$$

$$\boxed{U = \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{pmatrix}}$$

U- Matrix :

$$A_2 = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \xrightarrow{R1+2(R2)\to R2}$$

$$\begin{pmatrix} 2 & -1 & 0 \\ -1(2) & (2)(2) & -1(2) \\ 0 & -1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 & 0 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 & 0 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{pmatrix} \xrightarrow{R2+3R3\to R3}$$

$$\begin{pmatrix} 2 & -1 & 0 \\ 0 & 3 & -2 \\ 0(3) & -1(3) & (2)(3) \end{pmatrix} \begin{pmatrix} 2 & -1 & 0 \\ 0 & 3 & -2 \\ 0 & 0 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 & 0 \\ 0 & 3 & -2 \\ 0 & 0 & 4 \end{pmatrix} \begin{matrix} 1/2R_1 \to R1 \\ 1/3R2 \to R2 \\ 1/4R3 \to R3 \end{matrix} \longrightarrow$$

$$U = \begin{pmatrix} 1 & -1/2 & 0 \\ 0 & 1 & -2/3 \\ 0 & 0 & 1 \end{pmatrix}$$

L matrix :

$$A_2 = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

$R3 + 2R2 \rightarrow R2$

$$\begin{pmatrix} 2 & -1 & 0 \\ -1(2) & 2(2) & -1(2) \\ 0 & -1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & -1 & 0 \\ -2 & 3 & 0 \\ 0 & -1 & 2 \end{pmatrix}$$

$R2 + 3R1 \rightarrow R1$

$$= \begin{pmatrix} 2(3) & -1(3) & 0 \\ -2 & 3 & 0 \\ 0 & -1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 0 & 0 \\ -2 & 3 & 0 \\ 0 & -1 & 2 \end{pmatrix}$$

$\frac{1}{4}R1 \rightarrow R1$
$-\frac{1}{3}R2 \rightarrow R2$
$\frac{1}{2}R3 \rightarrow R3$

$$L_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2/3 & 1 & 0 \\ 0 & -1/2 & 1 \end{pmatrix}$$

L-matrix: $A_3 = \begin{pmatrix} 3 & 0 & 0 \\ -1 & 2 & 0 \\ 2 & 1 & 2 \end{pmatrix}$ $\frac{1}{3}R_1 \to R_1$, $\frac{1}{2}R_2 \to R_2$, $\frac{1}{2}R_3 \to R_3$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ 1 & 1/2 & 1 \end{pmatrix}$$

U-matrix:

$$A_3 = \begin{pmatrix} 3 & 0 & 0 \\ -1 & 2 & 0 \\ 2 & 1 & 2 \end{pmatrix} \quad 3R2 + R1 \to R2$$

$$\begin{pmatrix} 3 & 0 & 0 \\ -1(3) & 2(3) & 0(3) \\ 2 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 2 & 1 & 2 \end{pmatrix}$$

$-2R1 + 3R3 \to R3$

$$\begin{pmatrix} 3(-2) & 0(-2) & 0(-2) \\ 0 & 6 & 0 \\ 2(3) & 1(3) & 2(3) \end{pmatrix}$$

$$= \begin{pmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 3 & 6 \end{pmatrix} \xrightarrow{R2 - 2R3 \to R3}$$

$$= \begin{pmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & -12 \end{pmatrix}$$

$1/3\,R1 \to R1$
$1/6\,R2 \to R2$
$= 1/12\,R3 \to R3$

$$U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$